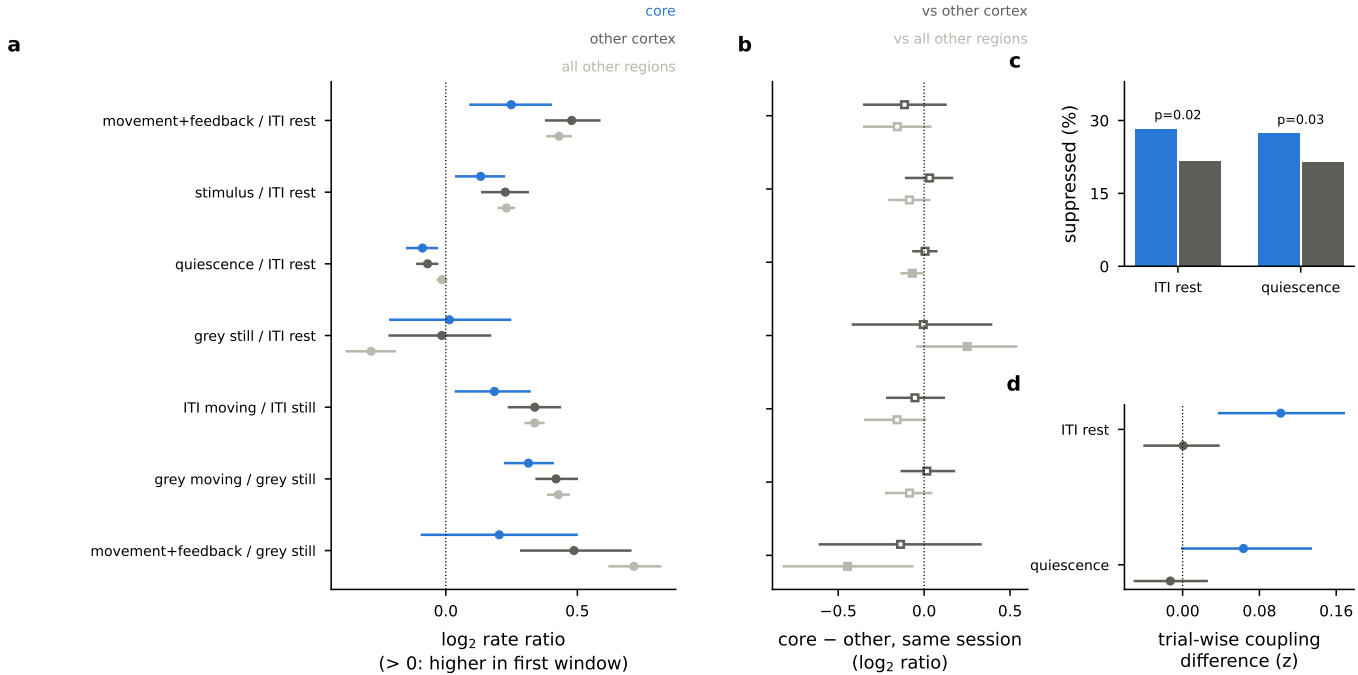


**Figure 6:** **a** DMN-motivated contrasts, per neuron $\log_2$ ratio of session-averaged rates (+0.1 Hz), mean over mice with 95% bootstrap CI, for the conservative core, all other cortex and all other recorded regions. A task-negative region should be below zero in the first three rows; the quiescence row separates engagement from movement, the movement rows separate movement from the task. **b** The most conservative version: core neurons minus non-core neurons recorded in the same session, averaged per mouse (31 mice vs other cortex, 49 vs all other regions); filled markers, Wilcoxon $p < 0.05$ over mice. **c** Fraction of neurons suppressed during movement + feedback relative to ITI rest or quiescence (per-trial Welch test, $p < 0.01$), core vs other cortex; region-permutation p. **d** Trial-by-trial co-fluctuation of region signals in ITI rest and quiescence windows (distance and probe residual; 39 core-core pairs, 26 mice): core-core minus core-other (blue) and core-other minus other-other (grey), 95% mouse-bootstrap CI. The core-core excess comes from same-probe neighbours ($<$0.5 mm, mainly RSPd–RSPv); distance-matched within session it is +0.06 (CI $-0.02$ to 0.13).